For, notational simplicity, we set

- $\hbar = c = 1$

In special relativity, our universe is represented by the vector space $\mathbb{R}^{1,3}$

- $\mathbb{R}^{1,3} := \mathbb{R}^4$ with a different inner product.

--- 이런식으로 ㄱㄱㄷ.

Leture 9 (special relativity)

- $\hbar = c = 1$ • Our universe $= \mathbb{R}^{1,3} = \mathbb{R}^4$ with a different inner product
- $x \in \mathbb{R}^{1,3}$. $x = (x^0, x^1, x^2, x^3)$, then x^0 is the time coordinate and $\mathbf{x} := (x^1, x^2, x^3)$ is the 3-tuple of spatial coordinates
- We define a symmetric matrix η as $\eta = \begin{pmatrix} 1 & & & 0 \\ & -1 & & \\ & & -1 & \\ 0 & & & -1 \end{pmatrix}$.
- $\eta_{\mu\nu} :=$ (μ,ν)-th entry of η
- Given $x, y \in \mathbb{R}^{1,3}$, $(x, y) = x^0y^0 - x^1y^1 - x^2y^2 - x^3y^3 = \sum_{\mu,\nu} \eta_{\mu\nu} x^\mu y^\nu$ (Minkowski 내적)
- $\mathbf{x} \cdot \mathbf{y} = x^1y^1 + x^2y^2 + x^3y^3$, thus $(x, y) = x^0y^0 - \mathbf{x}\cdot\mathbf{y}$
  standard 내적
- In special relativity, $x^2 = (x, x) = (x^0)^2 - \mathbf{x}\cdot\mathbf{x}$. • Euclidean norm of $\mathbf{x} := |\mathbf{x}|$

(위 노테이션에 익숙해지는 건 매우 중요하다!!!)

On the other hand, L (Lorentz transformation) on $\mathbb{R}^{1,3}$ is a linear map s.t. $(Lx, Ly) = (x, y)$ $\forall x, y \in \mathbb{R}^{1,3}$. $\Rightarrow L^T \eta L = \eta$ (이것은 usual inner product에서 orthogonality 조건과 같다)

The set of Lorentz transformations $= O(1,3)$
We have the following two homomorphisms w.r.t. L.

$$\begin{cases} \text{(i)} \ \det : O(1,3) \longrightarrow \mathbb{Z}_2 = \{-1, 1\} \\ \text{(ii)} \qquad L \longmapsto \operatorname{sign}(L^0_0) \text{, where } L^0_0 \text{ is the top left entry of the matrix} \end{cases}$$

$$SO^{\uparrow}(1,3) := \{ L \in O(1,3) \mid \det(L) = 1 \text{ and } \operatorname{sign}(L^0_0) = 1 \} \overset{\text{subgroup}}{\subset} O(1,3)$$

The parametrization of a curve in Euclidean space by arc length has the property that it is invariant under Euclidean symmetries. That is, under rotation and translation to the coordinate system, the parametrization remains unchanged.

~~As in Euclidean space~~, In a similar way to Euclidean space,

a curve in $\mathbb{R}^{1,3}$ can sometimes be parametrized in a way that is invariant under any change of the coordinate system by an action of $SO^{\uparrow}(1,3)$ or a spacetime translation.

Euclidean space vs Minkowski space

여기서 $\Leftarrow$ ([redacted] $\frac{dx}{d\tau}$ [redacted] $= 1$, [redacted] = Euclidean norm $\xleftarrow{\text{In a similar vein}}$ ✻ $\left(\frac{dx}{d\tau}, \frac{dx}{d\tau} \right)$, ( ) = Minkowski $= 1$

A sufficient condition is that in some coordinate system, the curve has a time parametrization $x(t) = (t, \mathbf{x}(t))$ satisfying $\left| \frac{d\mathbf{x}}{dt} \right| < 1$. $\forall t$

That is, in some coordinate system, the curve represents the trajectory of a particle whose speed is always strictly less than c.

<u>Def</u>: If $x(\tau)$ satisfies ✻, we call τ as 'proper time'.

- $x(\tau)$, τ: proper time. Mass of particle $= m$ then '<u>four-momentum</u>' at any point on the trajectory is $\underline{p = m \cdot \frac{dx}{d\tau}}$

※ Here, p does not depend on the coordinate system

- In a fixed coordinate system, the 'four-momentum' can be written

$$p = \left(\frac{m}{\sqrt{1 - v(t)^2}}, \frac{m v(t)}{\sqrt{1 - v(t)^2}} \right)$$, $v(t)$:= coordinate-dependent velocity

- $p = (p^0, p^1, p^2, p^3) = (p^0, \mathbf{p})$ $\longrightarrow$ (relativistic momentum)

$\longrightarrow$ (E, relativistic energy)

- $p^2 = (p^0)^2 - \mathbb{p}\cdot\mathbb{p} = m^2 \quad \therefore \quad p^0 = E = \sqrt{m^2+p^2} := w_p$

we suppress the dependence on the relativistic momentum

- The fourth momentum is always an element of the following manifold

$$X_m = \{p \in \mathbb{R}^{1,3} ; p^2 = m^2, \ p^0 \geq 0\}$$

This manifold is called the "mass shell" of mass m.

~~Lecture 10 (The mass~~

Lecture 10 (The mass shell)

- $H = L^2(X_m, d\lambda_m)$, where λ_m is the unique measure that is invariant under the action of the restricted Lorentz g.p on X_m

- How to define λ_m? [Luis, Miguel, Introduction Lectures on Quantum field theory]

Given an invariant function $f(p)$ of four momentum p of a particle mass m with $p^0 > 0$, $\exists$ integration measure

$$\int \frac{d^4p}{(2\pi)^4} (2\pi)\, \delta(p^2-m^2)\, \Theta(p^0)\, f(p) \sim (A)$$

manifestly Lorentz invariant for Lorentz transformation

where Θ is the Heaviside step fcn.

By using $\delta[f(x)] = \sum_{z_i = \text{zeros of } f} \frac{1}{|f'(x_i)|} \delta(x-x_i)$, we have

$$\delta(p^2-m^2) = \frac{1}{2p^0}\delta\left(p^0 - \sqrt{\mathbb{p}^2+m^2}\right) + \frac{1}{2p^0}\delta\left(p^0 + \sqrt{\mathbb{p}^2+m^2}\right)$$

$$\therefore (A) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\sqrt{\mathbb{p}^2+m^2}} f\left(\sqrt{\mathbb{p}^2+m^2}, \mathbb{p}\right)$$

So the relativistic invariant measure is given by

$$\int \frac{d^3p}{(2\pi)^3} \frac{1}{2w_p} \quad \text{with} \quad w_p := \sqrt{\mathbb{p}^2+m^2}.$$

(or, use $\delta[(x-a)(x-b)] = \frac{1}{|b-a|}\left[\delta(x-a)+\delta(x-b)\right]$)

Lecture 11 (The postulate of quantum field theory)

Q ~~What does it mean to say quantum state at a given time?~~

How to change the postulates in special relativity setting?

Recall that we had five postulates of quantum mechanics that were introduced in lec 2.

P1 ~~Note~~ Lorentz transforms changes the concepts of time slices and the notion of two events happening 'at the same time' need not remain

P2 the same under a change of coordinate system.

P3 The postulates need to be modified as below.

P4

P5

P_1 For any quantum system, $\exists$ H (Hilbert space) s.t the state of the system is described by vectors in H. A state is not for a given time but for all space time. A state gives a complete spacetime description of the system.

P_2 unchanged

P_3 unchanged

P_4 unchanged

P_5 There is no concept of the evolution of a state in QFT. This postulate is totally different from before.

$\boxed{\text{The preparatory concepts for } P_5}$

- The poincare g.p P is $\mathbb{R}^{1,3} \rtimes SO^{\uparrow}(1,3)$. $\mathbb{R}^{1,3}$ is the g.p of spacetime translations (for $x \in \mathbb{R}^{1,3}$, $x(y) = x+y$)

i.e., $P = \{(a,\Lambda) : a \in \mathbb{R}^{1,3}, \Lambda \in SO^{\uparrow}(1,3)\}$ with $(a,\Lambda)(b,\Omega) = (a+\Lambda b, \Lambda\Omega)$

This is the g.p of isometries of the Minkowski spacetime

- The action of $(a,\Lambda) \in P$ on $x \in \mathbb{R}^{1,3}$ is $(a,\Lambda)(x) = a + \Lambda x$
- U of poincare g.p in H is a map $U : P \to$ the space of unitary operators in H

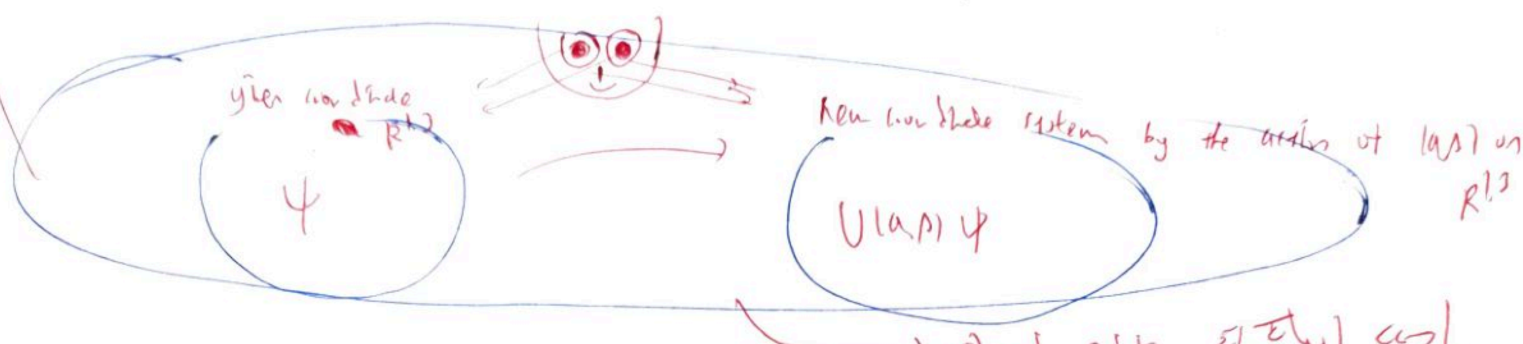
s.t $U((a,\Lambda)\cdot(b,\Omega)) = U(a,\Lambda)\, U(b,\Omega)$ $\forall$ $(a,\Lambda), (b,\Omega)$

- ~~Strongly continuous~~ We call the representation 'strongly continuous' if for any $\psi \in H$

$(a,\Lambda) \longmapsto U(a,\Lambda)\psi$ is continuous.

So, the revised P5 replaces the action of a time evolution g.p of unitary operators with a strongly continuous unitary representation of the poincare g.p

P5 Any quantum system with H comes equipped with a strongly continuous unitary representation U of P in H.



이건 그냥 [illegible]

Lecture 12 (The massive scalar field)

12-1 creation and annihilation on the mass shell.

- $H = L^2(X_m, d\lambda_m)$, $\mathcal{B}$ = Bosonic Fock space for this H.
- Recall A, A^+ on $\mathcal{B}$ and $\begin{cases} A(f) = \int d\lambda_m f^*(p)\, a(p) \\ A^+(f) = \int d\lambda_m f(p)\, a^+(p) \end{cases}$
- $(a(p)\phi)(p_1, \dots, p_n) = \sqrt{n}\, \phi(p_1, \dots, p_n, p)$, $(a^+(p)\phi)(p_1, \dots, p_{n+1}) = \frac{1}{\sqrt{n+1}} \sum_{j=1}^{n+1} \delta_{p_j}(p)\, \phi(p_1, \dots, \hat{p}_j, \dots, p_{n+1})$ — Dirac delta on X_m

※ We can not use $\delta(p - p_j)$ since X_m is not a vector space and so, $p - p_j$ may not belong to X_m.

- We define two related operator-valued distributions on $\mathcal{B}$.

$$a(\mathbb{p}) = \frac{1}{\sqrt{2w_{\mathbb{p}}}}\, a(p) \qquad a^+(\mathbb{p}) = \frac{1}{\sqrt{2w_{\mathbb{p}}}}\, a^+(p)$$

- commutation relations for $a(\mathbb{p})$ and $a^+(\mathbb{p})$

① $[a(\mathbb{p}), a(\mathbb{p}')] = 0$ ② $[a^+(\mathbb{p}), a^+(\mathbb{p}')] = 0$ ③ $[a(\mathbb{p}), a^+(\mathbb{p}')] = (2\pi)^3 \delta^{(3)}(\mathbb{p} - \mathbb{p}') \mathbb{1}$

To prove ③,

$$\begin{aligned} &\iint \frac{d^3\mathbb{p}}{(2\pi)^3} \frac{d^3\mathbb{p}'}{(2\pi)^3} f(\mathbb{p})^* g(\mathbb{p}') [a(\mathbb{p}), a^+(\mathbb{p}')] \\ &= \iint \frac{d^3\mathbb{p}}{(2\pi)^3} \frac{d^3\mathbb{p}'}{(2\pi)^3} \frac{1}{\sqrt{4w_{\mathbb{p}}w_{\mathbb{p}'}}} f(\mathbb{p})^* g(\mathbb{p}') [a(p), a^+(p')] \\ &= \iint d\lambda_m(p)\, d\lambda_m(p') \sqrt{4w_{\mathbb{p}}w_{\mathbb{p}'}}\, f(\mathbb{p})^* g(\mathbb{p}') \underbrace{[a(p), a^+(p')]}_{\delta_p(p') \cdot \mathbb{1}} \\ &= \int d\lambda_m(p)\, 2w_{\mathbb{p}}\, f(\mathbb{p})^* g(\mathbb{p}) \cdot \mathbb{1} = \left(\int \frac{d^3\mathbb{p}}{(2\pi)^3} f(\mathbb{p})^* g(\mathbb{p}) \right) \cdot \underbrace{\mathbb{1}}_{\text{The identity operator}} \end{aligned}$$

Now let me introduce the massive scalar free field.

- Φ (massive scalar field) is an operator-valued distributions ... of $\mathcal{S}(\mathbb{R}^{1,3})$ (Schwartz tensor on $\mathbb{R}^{1,3}$).
- The ~~action~~ of Φ on a $f \in \mathcal{S}(\mathbb{R}^{1,3})$ is ~~written~~ defined $\Phi(f) = A(\hat{f}^*|_{X_m}) + A^+(\hat{f}|_{X_m})$,

where $\hat{f}(p) = \int d^4x\, e^{i(x,p)} f(x)$.

Now, we move on to the Hamiltonian on Fock space. ~~the massive scalar boson~~

- $H :=$ the Hamiltonian for free evolution on $\mathcal{H} = L^2(X_m, d\lambda_m)$
- (Recall) on $L^2(\mathbb{R})$, $H = \int_{-\infty}^{\infty} dp \cdot \left(\frac{p^2}{2m}\right) a^+(p)\, a(p)$ (classical theory).

From the Hamiltonian for the evolution in the relativistic setting, $H_0\psi(p) = p^0\psi(p)$.

~~...~~ the above H_0 can be expressed by

12.3 Hamiltonian on Fock space

$$H_0 = \int_{X_m} d\lambda_m(p)\; p^0\, a^+(p)\cdot a(p)$$
$$= \int_{\mathbb{R}^3} \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{p}}}\, \omega_{\mathbf{p}} \sqrt{2\omega_{\mathbf{p}}}\, a^+(\mathbf{p}) \sqrt{2\omega_{\mathbf{p}}}\, a(\mathbf{p})$$
$$= \int_{\mathbb{R}^3} \frac{d^3\mathbf{p}}{(2\pi)^3}\, \omega_{\mathbf{p}}\, a^+(\mathbf{p})\cdot a(\mathbf{p})$$

12.4 Pointwise representation of the free field

- Φ (~~Real~~ Scalar free field of mass m), recall $f \in \mathcal{S}(\mathbb{R}^{1,3})$, $\hat{f}(p) = \int d^4x\, e^{i(x,p)} f(x)$ (A)

$$\Rightarrow \Phi(f) = A(\hat{f}^*|_{X_m}) + A^+(\hat{f}|_{X_m})$$
$$= \int d\lambda_m(p) \left(\hat{f}^*(p^*)\, a(p) + \hat{f}(p)\, a^+(p) \right)$$
$$= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \cdot \frac{1}{2\omega_{\mathbf{p}}} \left(\hat{f}^*(p)\, a(p) + \hat{f}(p)\, a^+(p) \right)$$
$$= \int_{\mathbb{R}^3} \frac{d^3\mathbf{p}}{(2\pi)^3} \cdot \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} \left(\hat{f}^*(p)\, a(\mathbf{p}) + \hat{f}(p) \cdot a^+(\mathbf{p}) \right)$$

plug (A) $\Rightarrow$ $$\Phi(f) = \int_{\mathbb{R}^{1,3}} d^4x\, f(x) \left[\underbrace{\int_{\mathbb{R}^3} \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} \left(e^{-i(x,p)} a(\mathbf{p}) + e^{i(x,p)} a^+(\mathbf{p}) \right)}_{=\,\Phi(x)} \right]$$

$\Phi(x)$. (pointwise representation of Φ).

12.5 Normal ordering

(5 page 참고하기)

For example, $a(\mathbf{p})\, a^+(\mathbf{p}) = \ldots a^+(\mathbf{p})\, a(\mathbf{p}) \ldots$

Normal ordering will play an important role in the contraction of interaction Hamiltonian later.

1.6 Expressing the Hamiltonian using the free fields

$x = (t, \mathbb{X})$ and $\mathbb{X} = (x^1, x^2, x^3)$.

Recall $\phi(x) = \int_{\mathbb{R}^3} \frac{d^3\mathbb{p}}{(2\pi)^3} \cdot \frac{1}{\sqrt{2\omega_{\mathbb{p}}}} \left(e^{-i(x,p)} a(\mathbb{p}) + e^{i(x,p)} a^\dagger(\mathbb{p}) \right)$.

$$\Rightarrow \quad H_0 = \int_{\mathbb{R}^3} \frac{d^3\mathbb{p}}{(2\pi)^3} \omega_{\mathbb{p}}\, a^\dagger(\mathbb{p}) \cdot a(\mathbb{p}) = \frac{1}{2}\int_{\mathbb{R}^3} d^3\mathbb{X} :\left((\partial_t \phi(x))^2 + \sum_{\nu=1}^{3} (\partial_\nu \phi(x))^2 + m^2 \phi(x)^2 \right):$$

pf)
$$\text{RHS} = \frac{1}{2}\int \frac{d^3\mathbb{X}\, d^3\mathbb{p}\, d^3\mathbb{q}}{(2\pi)^6} :\Big[-\frac{\sqrt{\omega_{\mathbb{p}}\omega_{\mathbb{q}}}}{2} \left(e^{i(x,p)} a(\mathbb{p}) - e^{-i(x,p)} a^\dagger(\mathbb{p})\right)\left(e^{i(x,q)} a(\mathbb{q}) - e^{-i(x,q)} a^\dagger(\mathbb{q})\right)$$
$$+ \frac{1}{2\sqrt{\omega_{\mathbb{p}}\omega_{\mathbb{q}}}} \left(i\mathbb{p}\, e^{i(x,p)} a(\mathbb{p}) - i\mathbb{p}\, e^{-i(x,p)} a^\dagger(\mathbb{p}) \right)\cdot\left(i\mathbb{q}\, e^{i(x,q)} a(\mathbb{q}) - i\mathbb{q}\, e^{-i(x,q)} a^\dagger(\mathbb{q}) \right)$$
$$+ \frac{m^2}{2\sqrt{\omega_{\mathbb{p}}\omega_{\mathbb{q}}}} \left(e^{i(x,p)} a(\mathbb{p}) + e^{-i(x,p)} a^\dagger(\mathbb{p}) \right)\left(e^{i(x,q)} a(\mathbb{q}) + e^{-i(x,q)} a^\dagger(\mathbb{q}) \right) \Big]:$$
$$= \frac{1}{4}\int \frac{d^3\mathbb{p}}{(2\pi)^3} \frac{1}{\omega_{\mathbb{p}}} :\Big[\left(-\omega_{\mathbb{p}}^2 + \mathbb{p}^2 + m^2\right) \cdot \left(a(\mathbb{p}) a^\dagger(\mathbb{q}) + a^\dagger(\mathbb{p}) \cdot a(\mathbb{q}) \right) +$$
$$\left(\omega_{\mathbb{p}}^2 + \underbrace{m^2 + \mathbb{p}^2}_{\omega_{\mathbb{p}}^2} \right) \cdot \left(a(\mathbb{p}) \cdot a^\dagger(\mathbb{p}) + a^\dagger(\mathbb{p}) \cdot a(\mathbb{p}) \right) \Big]:$$
$$= \frac{1}{4}\int \frac{d^3\mathbb{p}}{(2\pi)^3} \cdot \omega_{\mathbb{p}} :\left(a^\dagger(\mathbb{p}) \cdot a(\mathbb{p}) + a(\mathbb{p})\, a^\dagger(\mathbb{p}) \right):$$
$$= \frac{1}{2}\int \frac{d^3\mathbb{p}}{(2\pi)^3} \cdot \omega_{\mathbb{p}} \cdot \left(a^\dagger(\mathbb{p})\, a(\mathbb{p}) \right)$$

$\square$

-X. Here, from the second to the third, we use the relation

$$\left(\begin{aligned} \int d^3\mathbb{X} \cdot e^{i(x,p)} &= (2\pi)^3 \cdot \delta^{(3)}(\mathbb{p}) \\ \int d^3\mathbb{X} \cdot e^{i(x,q)} &= (2\pi)^3 \cdot \delta^{(3)}(\mathbb{q}) \end{aligned} \right)$$

to integrate it w.r.t $d^3\mathbb{X}$.